

DTSA 5745

Asymmetric Cryptography and Key Management

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (48.7 / 100)	41.3 / 100
SPECIALIZATION Applied Cryptography		

41

CURRENT AVG SCORE
across 6 job roles

74

POTENTIAL SCORE
with recommended additions

+48

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course will teach the principles of asymmetric cryptography (a.k.a. public-key cryptography) and describe how the key-pair use can enable different security properties/applications. We will study the popular ciphers, e.g., RSA and the Diffie-Hellman Exchange, and learn how/why they work to secure networking, including the mathematical problems anchoring their security. We will learn about the key distribution and management and digital certificates. Required for this course is a working knowledge of discrete mathematics.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	48.7	79.5	+30.8	HIGH
Machine Learning Engineer	41.5	71.1	+29.6	MEDIUM
Data Scientist	41.2	89.5	+48.2	HIGH
Applied Scientist	40.9	84.2	+43.3	HIGH
ML Engineer	38.8	62.9	+24.1	MEDIUM
MLOps Engineer	36.9	58.9	+22.0	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

LOW

Optional: Feature Engineering — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

AI Engineer

48.7 → 79.5 (+30.8 pts)

data models learning build software technical code systems

Machine Learning Engineer

41.5 → 71.1 (+29.6 pts)

data learning tools protection development science models code

Data Scientist

41.2 → 89.5 (+48.2 pts)

definition data product mitigation build approaches

Applied Scientist

40.9 → 84.2 (+43.3 pts)

computational interdisciplinary models technical energy data learning systems

ML Engineer

38.8 → 62.9 (+24.1 pts)

data learning machine users systems models build product

MLOps Engineer

36.9 → 58.9 (+22.0 pts)

data systems learning machine preprocessing tools models development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.